

CH7015 SDTV / HDTV Encoder

Features

- VGA to SDTV conversion supporting graphics resolutions up to 1024x768
- Analog YPrPb output for HDTV
- HDTV support for 480p, 576p, 720p, 1080i and 1080p
- Macrovision™ 7.1.L1 copy protection support
- Programmable digital input interface supporting RGB and YCrCb input data formats
- True scale rendering engine supports under-scan in all TV output resolutions
- Text enhancement filter
- Adaptive flicker filter with up to 7 lines of filtering
- Interlaced to progressive scan conversion for DVD
- Support for NTSC, PAL and HDTV formats
- Support for SCART connector
- Outputs CVBS, S-Video, RGB and YPrPb
- Support for Wide Screen Signaling (WSS)
- TV / Monitor connection detect
- Programmable power management
- Four 10-bit video DAC outputs
- Fully programmable through serial port
- Complete Windows and DOS driver support
- Low voltage interface support to graphics device
- Offered in a 48-pin LQFP package

1.0 GENERAL DESCRIPTION

The CH7015 is a Display Controller device which accepts a digital graphics input signal, and encodes and transmits data through a 10-bit high speed DAC. The device is able to encode the video signals and generate synchronization signals for NTSC and PAL TV standards (SDTV), as well as analog HDTV interface standards and graphics standards up to UXGA. The device accepts data over one 12-bit wide variable voltage data port which supports 5 different data formats including RGB and YCrCb.

The TV-Out processor will perform non-interlace to interlace conversion with scaling and flicker filter, and encode data into any of the NTSC or PAL video standards. The scaling and flicker filter is adaptive and programmable to enable superior text display. Eight graphics resolutions are supported up to 1024 by 768 with full vertical and horizontal under-scan capability in all modes. A high accuracy low jitter phase locked loop is integrated to create outstanding video quality. Support is provided for Macrovision™. ITU-R BT.656 interlaced video can also be input and scan converted to non-interlaced video.

In addition to TV encoder modes, bypass modes are included which perform color space conversion to HDTV standards and generate and insert HDTV sync signals, or output VGA style analog RGB for use as a CRT DAC.

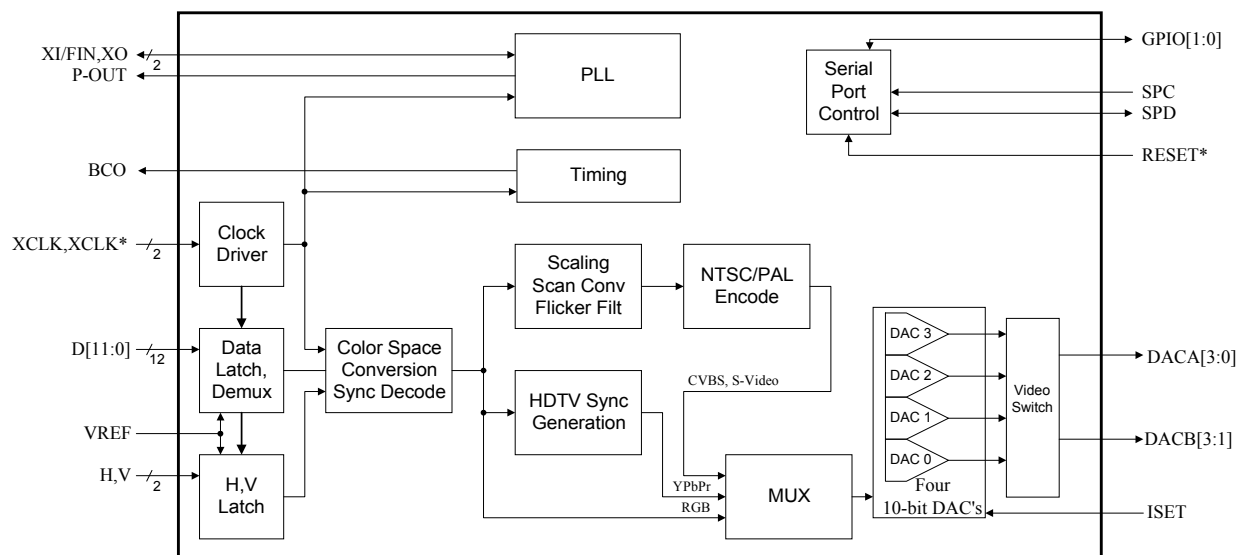


Figure 1: Functional Block Diagram

2.0 PIN-OUT

2.1 Package Diagram

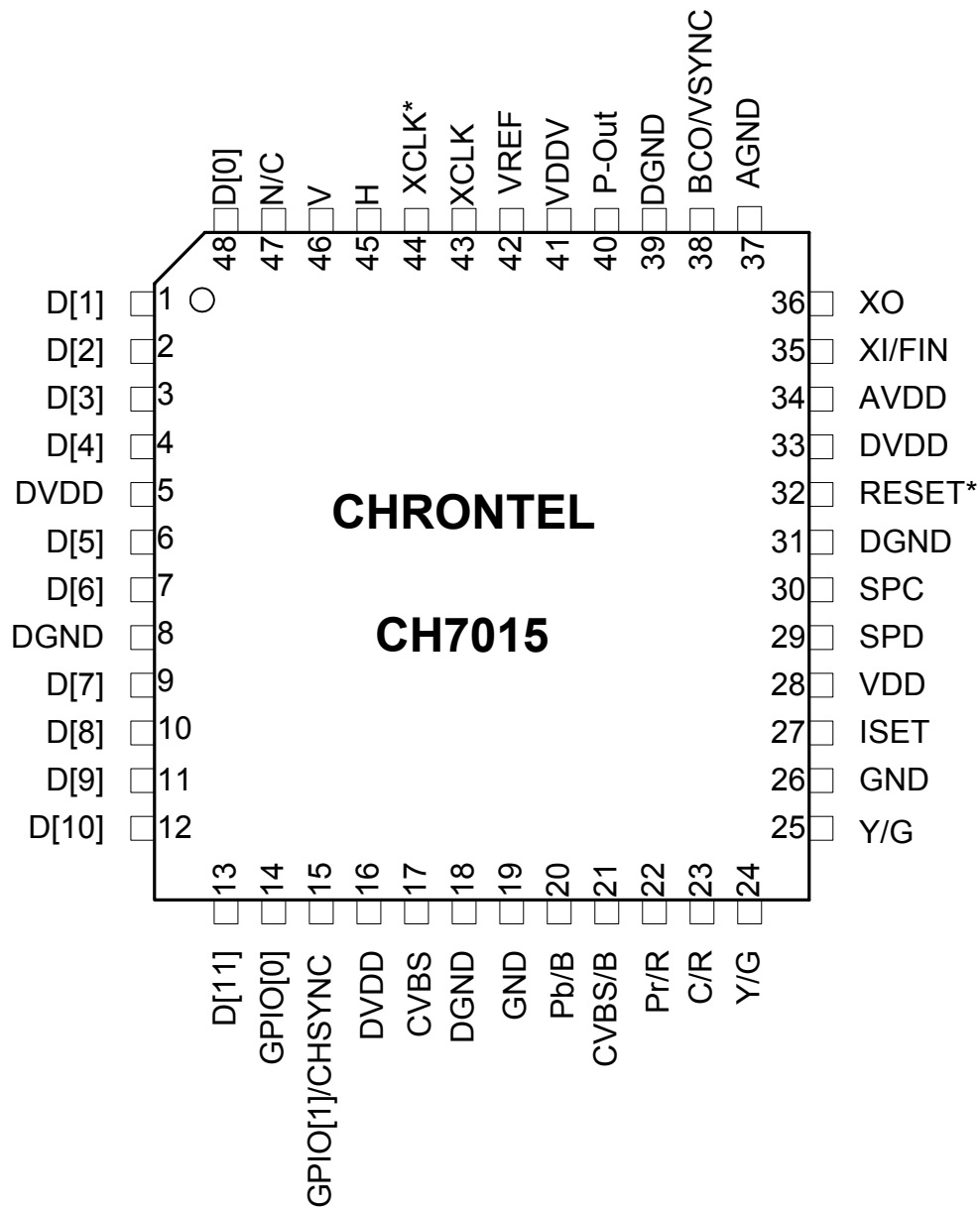


Figure 2: 48-Pin LQFP Package

2.2 Pin Description

Table 1: Pin Description

Pin #	Type	Symbol	Description
1-4, 6,7, 9-13, 48	In	D[11]-D[0]	Data[11] through Data[0] Inputs These pins accept the 12 data inputs from a digital video port of a graphics controller. The levels are 0 to VDDV, and the VREF signal is used as the threshold level.
14	In/Out	GPIO0	General Purpose Input – Output0 (weak internal pull-up) This pin provides general purpose I/O controlled via the serial port. This allows an external switch to be used to select NTSC or PAL at power-up. The internal pull-up will be to the DVDD supply.
15	In/Out	GPIO1 / CHSYNC	General Purpose Input – Output1 (weak internal pull-up) This pin provides general purpose I/O controlled via the serial port. This allows an external switch to be used to select NTSC or PAL at power-up. The internal pull-up will be to the DVDD supply. It can also be configured to output composite or horizontal sync.
17	Out	CVBS (DAC3)	Composite Video This pin outputs a composite video signal capable of driving a 75 ohm doubly terminated load. During bypass modes this output is valid only if the data format is compatible with one of the TV-Out display modes.
20	Out	Pb/B (DACB0)	Pb / Blue Output This pin outputs a selectable video signal. The output is designed to drive a 75 ohm doubly terminated load. The output can be selected to the Pb component of YPrPb or blue (for VGA bypass).
21	Out	CVBS/B (DACA0)	Composite Video / Blue Output This pin outputs a selectable video signal. The output is designed to drive a 75 ohm doubly terminated load. The output can be selected to be composite video or blue (for SCART type 1 connections).
22	Out	Pr/R (DACB2)	Pr / Red Output This pin outputs a selectable video signal. The output is designed to drive a 75 ohm doubly terminated load. The output can be selected to be the Pr component of YPrPb or red (for VGA bypass).
23	Out	C/R (DACA2)	Chroma / Red Output This pin outputs a selectable video signal. The output is designed to drive a 75 ohm doubly terminated load. The output can be selected to be s-video chrominance or red (for SCART type 1 connections).
24	Out	Y/G (DACB1)	Luma / Green Output This pin outputs a selectable video signal. The output is designed to drive a 75 ohm doubly terminated load. The output can be selected to be the luminance component of YPrPb or green (for VGA bypass).
25	Out	Y/G (DACA1)	Luma / Green Output This pin outputs a selectable video signal. The output is designed to drive a 75 ohm doubly terminated load. The output can be selected to be s-video luminance or green (for SCART type 1 connections).
27	In	ISSET	Current Set Resistor Input This pin sets the DAC current. A 140 ohm resistor should be connected between this pin and GND (pin 24 or 26) using short and wide traces.
29	In/Out	SPD	Serial Port Data Input / Output This pin functions as the bi-directional data pin of the serial port and operates with inputs from 0 to VDDV. Outputs are driven from 0 to VDDV. The serial port addresses for the CH7015 and CH7205 are 75h and 76h respectively.
30	In	SPC	Serial Port Clock Input This pin functions as the clock pin of the serial port and operates with inputs from 0 to VDDV.
32	In	RESET*	Reset * Input (Internal pull-up) When this pin is low, the device is held in the power-on reset condition. When this pin is high, reset is controlled through the serial port.
35	In	XI / FIN	Crystal Input / External Reference Input A parallel resonance 14.31818MHz crystal (± 20 ppm) should be attached between this pin and XO. However, an external CMOS compatible clock can drive the XI/FIN input.

Table 1: Pin Description (continued)

Pin #	Type	Symbol	Description
36	Out	XO	Crystal Output A parallel resonance 14.31818MHz crystal (± 20 ppm) should be attached between this pin and XI / FIN. However, if an external CMOS clock is attached to XI/FIN, XO should be left open.
38	Out	BCO	Buffered Clock Output This output pin provides selectable buffered clocks to be output, driven by the DVDD supply. The output clock can be selected using the BCO register. The levels are 0 to DVDD.
40	Out	P-Out	Pixel Clock Output This pin provides a pixel clock signal to the VGA controller which can be used as a reference frequency. The output is selectable between 1X and 2X of the pixel clock frequency. The output driver is driven from the VDDV supply. This output has a programmable tri-state. The capacitive loading on this pin should be kept to a minimum.
42	In	VREF	Reference Voltage Input The VREF pin inputs a reference voltage of $VDDV / 2$. The signal is derived externally through a resistor divider and decoupling capacitor, and will be used as a reference level for data, sync and clock inputs.
43, 44	In	XCLK, XCLK*	External Clock Inputs These inputs form a differential clock signal input to the device for use with the H, V and D[11:0] data. If differential clocks are not available, the XCLK* input should be connected to VREF. The clock polarity used can be selected by the MCP control bit.
45	In/Out	H	Horizontal Sync Input / Output When the SYO control bit is low, this pin accepts a horizontal sync input for use with the input data. The amplitude will be 0 to VDDV, and the VREF signal is used as the threshold level. This pin must be used as an input in all bypass modes. When the SYO control bit is high, the device will output a horizontal sync pulse, 64 pixels wide. The output is driven from the DVDD supply. This output is valid with TV-Out operation.
46	In/Out	V	Vertical Sync Input / Output When the SYO control bit is low, this pin accepts a vertical sync input for use with the input data. The amplitude will be 0 to VDDV, and the VREF signal is used as the threshold level. This pin must be used as an input in all bypass modes. When the SYO control bit is high, the device will output a vertical sync pulse one line wide. The output is driven from the DVDD supply. This output is valid with TV-Out operation.
5,16,33	Power	DVDD	Digital Supply Voltage (3.3V)
8,18,31,39	Power	DGND	Digital Ground
41	Power	VDDV	I/O Supply Voltage (1.1V to 3.3V)
34	Power	AVDD	PLL Supply Voltage (3.3V)
37	Power	AGND	PLL Ground
28	Power	VDD	DAC Supply Voltage (3.3V)
19,26	Power	GND	DAC Ground

3.0 FUNCTIONAL DESCRIPTION

3.1 Modes of Operation

The CH7015 is capable of being operated as a VGA to TV encoder, an ITU-R BT.601/656 encoder (with or without a de-interlacing function), or in one of several bypass modes for driving monitors requiring component video signals (HDTV, multi-sync monitors, etc.). All modes make use of the same set of DAC's, and therefore cannot be used simultaneously. Table 2 describes the possible operating modes. A 'p' following a number in the Input Scan Type column indicates a progressive scan (non-interlaced) input where the number indicates the active number of lines per frame. An 'i' following a number in the Input Scan Type column indicates an interlaced input where the number indicates the active number of lines per frame. Detailed descriptions of each of the operating modes follows Table 2.

Table 2: Operating Modes

Input Scan Type	Input Data Format	Output scan Type	Output Format	Operating Mode	Described In section
non-interlaced	RGB	non-interlaced	RGB	RGB bypass	3.1.5
non-interlaced (480p, 576p, 720p)	RGB / YCrCb ¹	non-interlaced	YPbPr ^{2,3}	HDTV/EDTV bypass	3.1.4
non-interlaced (VGA -> XGA)	RGB / YCrCb ¹	Interlaced	CVBS, S-Video	VGA to SDTV encoder (NTSC / PAL)	3.1.1
non-interlaced (VGA -> XGA)	RGB / YCrCb ¹	Interlaced	CVBS, RGB	VGA to SDTV encoder (SCART format)	3.1.1
Interlaced (1080i)	RGB / YCrCb ¹	Interlaced	YPbPr ³	HDTV/EDTV bypass (1080i)	3.1.4
non-interlaced (1080p)	RGB / YCrCb ¹	non-interlaced	YPbPr ³	HDTV/EDTV bypass (1080p)	3.1.4
interlaced (480i, 576i)	RGB / YCrCb ¹	non-interlaced	YPbPr ²	ITU-R BT.601/656 TV Deinterlace (480p, 576p generation)	3.1.3
interlaced (480i, 576i)	RGB / YCrCb ¹	Interlaced	CVBS, S-Video	ITU-R BT.601/656 TV Encoder (NTSC / PAL)	3.1.2
interlaced (480i, 576i)	RGB / YCrCb ¹	Interlaced	CVBS, RGB	ITU-R BT.601/656 TV Encoder (SCART format)	3.1.2
interlaced (480i, 576i)	RGB / YCrCb ¹	Interlaced	YPbPr ²	ITU-R BT.601/656 TV Encoder	3.1.2

¹ YCrCb signal has the following characteristics (assumed to be gamma corrected):

$$Y = 77/256 * R + 150/256 * G + 29/256 * B$$

$$Cr = 131/256 * R - 110/256 * G - 21/256 * B + 128$$

$$Cb = -44/256 * R - 87/256 * G + 131/256 * B + 128$$

Data is 8-bit wide ITU-R BT.656 format

Data Sequence is Cb₀Y₀Cr₀Y₁Cb₂Y₂Cr₂Y₃

where Cb₀Y₀Cr₀ are co-sited samples, and Y₁ is the following luma sample

² YPrPb signal has the following characteristics (according to SMPTE 293M):

$$Y = 0.299 * R + 0.587 * G + 0.114 * B$$

$$Pb = 0.56433 * (B - Y)$$

$$Pr = 0.71327 * (R - Y)$$

Bi / Tri level sync inserted on each analog component

³ YPrPb signal has the following characteristics (according to SMPTE 274M, 295M, 296M):

$$Y = 0.2126 * R + 0.7154 * G + 0.0722 * B$$

$$Pb = 0.53891 * (B - Y)$$

$$Pr = 0.63500 * (R - Y)$$

Bi / Tri level sync inserted on each analog component

3.1.1 VGA to SDTV Encoder

In VGA to SDTV Encoder mode non-interlaced data, sync and clock signals are input to the device from a graphics controllers digital output port. A clock signal (P-Out) can be output as a frequency reference to the graphics controller, which is recommended to ensure accurate frequency generation. Horizontal and vertical sync signals are normally sent to the device from the graphics controller, but can be embedded into the data stream in YCrCb input data formats, or can be output to the graphics controller. Data is 2X multiplexed, and the XCLK clock signal can be 1X or 2X times the pixel rate. Input data will be scaled, scan converted and flicker filtered, then encoded into the selected video standard and output from the video DAC's. NTSC and PAL formats are supported. The device can output data in S-Video and CVBS format, or as RGB for interface to a SCART connector. The graphics resolutions supported for VGA to TV output are shown in Table 3 below. As the table shows, the amount of scaling is programmable. The flicker filter includes text enhancement circuitry, and the amount of flicker filtering and text enhancement performed is programmable, with the maximum number of taps used in the filter being five, six or seven depending upon the scaling ratio. The device is capable of adding Macrovision encoding to the output signal.

Table 3: VGA to TV Operating Modes

Input Resolution	Active Video Aspect Ratio	Pixel Aspect Ratio	TV Output Standard	Scaling Ratios
512x384	4:3	1:1	PAL	5/4, 1/1
512x384	4:3	1:1	NTSC	5/4, 1/1
720x400	4:3	1.35:1.00	PAL	5/4, 1/1
720x400	4:3	1.35:1.00	NTSC	5/4, 1/1, 25/21
640x400	8:5	1:1	PAL	5/4, 1/1
640x400	8:5	1:1	NTSC	5/4, 1/1, 7/8, 25/21
640x480	4:3	1:1	PAL	5/4, 1/1, 5/6, 25/21
640x480	4:3	1:1	NTSC	1/1, 7/8, 5/6
720x480	4:3	9:8	NTSC	1/1, 7/8, 5/6
720x576	4:3	15:12	PAL	1/1, 5/6, 5/7
800x600	4:3	1:1	PAL	1/1, 5/6, 5/7
800x600	4:3	1:1	NTSC	3/4, 7/10, 5/8
1024x768	4:3	1:1	PAL	5/7, 5/8, 5/9
1024x768	4:3	1:1	NTSC	5/8, 5/9, 1/2

3.1.2 ITU-R BT.601/656 TV Encoder

In ITU-R BT.601/656 TV Encoder mode, interlaced data, sync and clock signals are input to the CH7015 from a graphics controllers digital output port, or the output of an MPEG decoder device. The YCrCb data format is most commonly used in these modes, but RGB data can be used as well. A clock signal (P-Out) can be output as a frequency reference to the graphics device. Horizontal and vertical sync signals are normally sent to the CH7015 from the graphics device, but can be embedded into the data stream in YCrCb input data formats, or can be output to the graphics controller. Data is 2X multiplexed, and the XCLK clock signal is 2X times the pixel rate. Input data bypasses the scaling, scan conversion and flicker filtering blocks, is encoded into the selected video standard and output from the video DAC's. NTSC and PAL formats are supported. The device can output data in S-Video and CVBS format, or as RGB for interface to a SCART connector or in YPrPb format. The graphics resolutions supported for ITU-R BT.601/656 TV output are shown in Table 4 below. The device is capable of adding Macrovision encoding to the output signal. The timing of the sync signals is shown in Figure 3 below. Note that the alignment of the VSYNC signal to the HSYNC signal changes from field 1 to field 2 to allow the CH7015 to identify the correct field.

Table 4: ITU-R BT.601/656 TV Encoder Operating Modes

Input Resolution	Active Video Aspect Ratio	Pixel Aspect Ratio	TV Output Standard	Scaling Ratios
720x480i	4:3	9:8	NTSC	1/1
720x576i	4:3	15:12	PAL	1/1

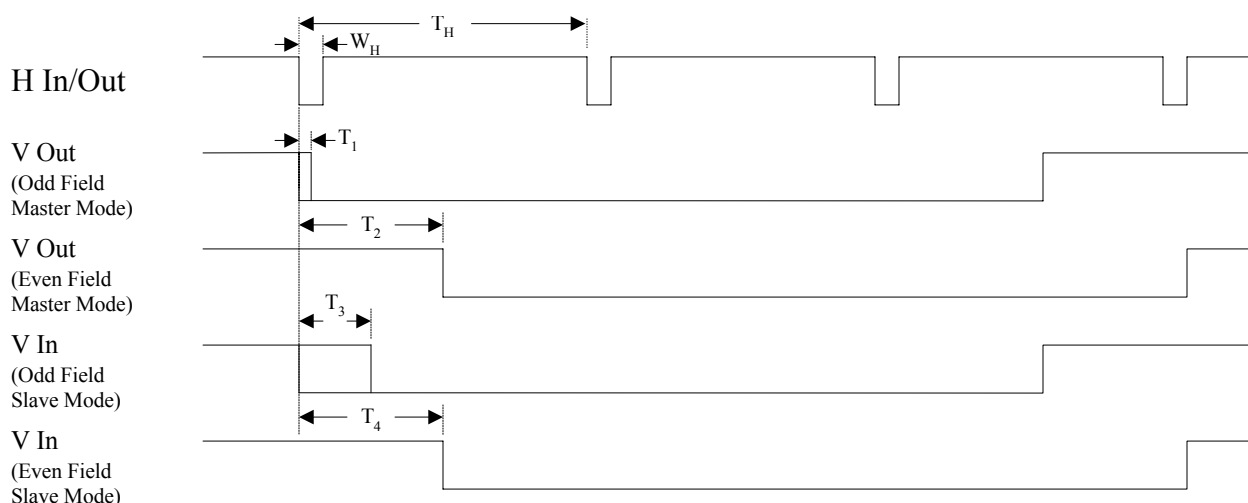


Figure 3: Interlaced Sync Input/Output Timing

Table 5: Interlaced Sync Input/Output Timing

Symbol	Parameter	Min	Typ	Max	Unit
T_{PCK}	Input clock period	6.73		47.62	uS
T_H	Total Line Period				
	SDTV	63.5		63.5	uS
	HDTV	14.8		37.0	uS
W_H	Hsync Width				
	When output from CH7015	64		64	Pixel clocks
T_1	When input to CH7015	1			Pixel clocks
T_1	Odd Field (Field 1) V SYNC out to H SYNC out alignment		0		uS
T_2	Even Field (Field 2) V SYNC out delay from H SYNC out		$0.5 * T_H$		uS
T_3	Odd Field (Field 1) V SYNC in to H SYNC in alignment	0		$W_H - T_{PCK}$	uS
T_4	Even Field (Field 2) V SYNC in delay from H SYNC in	W_H		$T_H - T_{PCK}$	uS

3.1.3 ITU-R BT.601/656 TV De-Interlace

In ITU-R BT.601/656 TV De-Interlace mode, interlaced data, sync and clock signals are input to the CH7015 from a graphics controllers digital output port, or the output of an MPEG decoder device. The YCrCb data format is most commonly used in these modes, but RGB data can be used as well. A clock signal (P-Out) can be output as a frequency reference to the graphics device. Horizontal and vertical sync signals are normally sent to the CH7015 from the graphics device, but can be embedded into the data stream in YCrCb input data formats, or can be output to the graphics controller. Data is 2X multiplexed, and the XCLK clock signal is 2X times the pixel rate. Input data is scan converted from interlaced to non-interlaced data, color space converted to the selected video format, has sync signals generated and is output from the video DAC's. The output format is YPbPr. The graphics resolutions supported for ITU-R BT.601/656 TV De-Interlace mode are shown in Table 6 below. The timing of the sync signals is shown in Figure 3 above. Note that the alignment of the VSYNC signal to the HSYNC signal changes from field 1 to field 2 to allow the CH7015 to identify the correct field.

Table 6: ITU-R BT.601/656 TV De-Interlace Operating Modes

Input Resolution	Active Video Aspect Ratio	Pixel Aspect Ratio	TV Output Standard	Scaling Ratios
720x480i	4:3	9:8	720x480p	1/1
720x576i	4:3	15:12	720x576p	1/1

3.1.4 HDTV / EDTV Bypass

In HDTV / EDTV Bypass mode, data, sync and clock signals are input to the CH7015 from a graphics device in the scanning method that matches the display device (interlaced data is sent to the CH7015 to drive an interlaced display, non-interlaced data is sent to the CH7015 to drive a non-interlaced display). The input data format can be YCrCb or RGB. A clock signal (P-Out) can be output as a frequency reference to the graphics device. Horizontal and vertical sync signals must either be sent to the CH7015 from the graphics device or embedded in the data stream according to SMPTE standards. Data is 2X multiplexed, and the XCLK clock signal can be 1X or 2X times the pixel rate. Input data is color space converted to the selected video format, has sync signals generated and is output from the video DAC's. The output format is YPbPr. The graphics resolutions supported for HDTV Bypass mode are shown in Table 7 below. The resolutions supported for EDTV Bypass mode are shown in Table 8: below. The timing of the sync signals for interlaced input modes is shown in Figure 3 above. Note that in interlaced input modes the alignment of the VSYNC signal to the HSYNC signal changes from field 1 to field 2 to allow the CH7015 to identify the correct field. No scaling, scan conversion or flicker filtering is applied in HDTV / EDTV Bypass mode.

Table 7: HDTV Bypass

Input Resolution	Total Resolution	Scan Type	Pixel Clock (MHz)	Frame Rate (Hz)	Standard
1280x720	1650x750	Non-Interlaced	74.25 74.25/1.001	60 60/1.001	SMPTE 296M
1920x1080	2200x1125	Interlaced	74.25 74.25/1.001	30 30/1.001	SMPTE 274M
1920x1080	2640x1125	Interlaced	74.25	25	SMPTE 274M
1920x1080	2376x1250	Interlaced	74.25	25	SMPTE 295M
1920x1080	2200x1125	Non-Interlaced	148.5 148.5/1.001 74.25 74.25/1.001	60 60/1.001 30 30/1.001	SMPTE 274M
1920x1080	2640x1125	Non-Interlaced	148.5 74.25	50 25	SMPTE 274M
1920x1080	2750x1125	Non-Interlaced	74.25 74.25/1.001	24 24/1.001	SMPTE 274M
1920x1080	2376x1250	Non-Interlaced	148.5	50	SMPTE 295M

Table 8: EDTV Bypass

Active Resolution	Total Resolution	Scan Type	Pixel Clock (MHz)	Frame Rate (Hz)	Standard
720x480	858x525	Non-Interlaced	27.0	60/1.001	EIA-770.2-A
720x483	858x525	Non-Interlaced	27.027	60	SMPTE 293M
720x576	864x625	Non-interlaced	27.0	50	ITU-R BT.1358

3.1.5 RGB Bypass

In RGB Bypass mode, data, sync and clock signals are input to the CH7015 from a graphics device, and bypassed directly to the D/A converters to implement a second CRT DAC function. External sync signals must be supplied from the graphics device. These sync signals are buffered internally, and can be output to drive the CRT. The input data format must be RGB in this operating mode. Input data is 2X multiplexed, and the XCLK clock signal can be 1X or 2X times the pixel rate. The CH7015 can support a pixel rate of 165MHz. This operating mode uses 8-bits of three of the DAC's 10-bit range, and provides a nominal signal swing of 0.661V (or 0.7V depending on DAC Gain setting in control registers) when driving a 75Ω doubly terminated load. No scaling, scan conversion or flicker filtering is applied in RGB Bypass modes.

3.2 TV Encoder / Bypass RGB / Component Video Outputs

Three of the four TV encoder DAC outputs can be switched to two sets of output pins DACA[2:0] and DACB[2:0] via video switches. The fourth DAC output, DAC3 is not switched so that CVBS output is available for the HDTV configuration in addition to the standard TV output configurations. This feature facilitates simple connection to two sets of video connectors as listed in Table 9.

Table 9: TV Output Configurations

Pin # (name)	2 RCA + 1 S-Video	SCART	
Pin 21 (DACA0)	CVBS	B	
Pin 25 (DACA1)	Y	G	
Pin 23 (DACA2)	C	R	
Pin 17 (DAC3)	CVBS	CVBS	
	VGA – Bypass RGB	HDTV	SDTV ¹
Pin 20 (DACB0)	B	Pb	Pb
Pin 24 (DACB1)	G	Y	Y
Pin 22 (DACB2)	R	Pr	Pr
Pin 17 (DAC3)			CVBS

If the application calls for CVBS/S-video, SCART, RGB and YPrPb to output on one set of DAC output pins, different reconstruction filters for each type of signal can be implemented on the break-out cables.

The TV Encoder can be bypassed and input data drives the DACs directly. This mode can go to 165MP/s. The CH7015 supports YPrPb output for driving 480i, 480p, 576i, 576p, 720p, 1080i and 1080p TV sets, and SCART RGB for European TV.

Note: CVBS is available with YPrPb for SDTV interlaced outputs.

3.3 Input Interface

3.3.1 Overview

Two distinct methods of transferring data to the CH7015 are described. They are:

- Multiplexed data, clock input at 1X the pixel rate
- Multiplexed data, clock input at 2X the pixel rate

For the multiplexed data, clock at 1X pixel rate, the data applied to the CH7015 is latched with both edges of the clock (also referred to as dual edge transfer mode or DDR). For the multiplexed data, clock at 2X pixel rate the data applied to the CH7015 is latched with one edge of the clock (also known as single edge transfer mode or SDR). The polarity of the pixel clock can be reversed under serial port control. In single edge transfer modes, the clock edge used to latch data is programmable. In dual edge transfer modes, the clock edge used to latch the first half of each pixel is programmable.

3.3.2 Input Clock and Data Timing Diagram

Figure 4 below shows the timing diagram for input data and clocks. The first XCLK/XCLK* waveform represents the input clock for single edge transfer (SDR) methods. The second XCLK/XCLK* waveform represents the input clock for the dual edge transfer (DDR) method. The timing requirements are given in section 4.5.

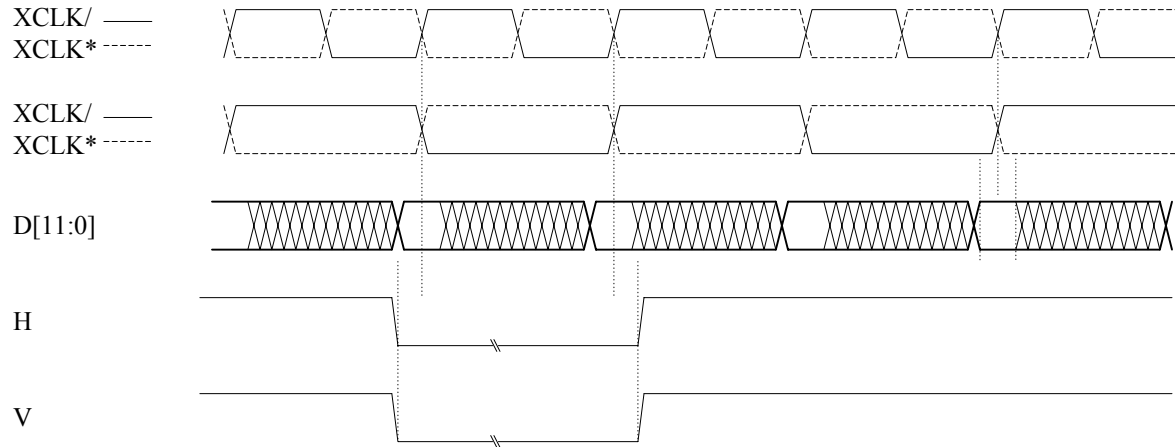


Figure 4: Clock, Data and Interface Timing

3.3.3 Data De-skew Feature

The de-skew feature allows adjustment of the input setup and hold time. The input data D[11:0] can be latched slightly before or after the latching edge of XCLK depending on the amount of the de-skew. Note that the XCLK is not changed, only the time at which the data is latch relative to XCLK. The de-skew is controlled using the XCMD[3:0] bits located in register 1Dh. The delay t_{CD} between clock and data is given by the following formula:

$$t_{CD} = -XCMD[3:0] * t_{STEP} \text{ for } 0 \leq XCMD[3:0] \leq 7$$

$$t_{CD} = (XCMD[3:0] - 8) * t_{STEP} \text{ for } 8 \leq XCMD[3:0] \leq 15$$

where XCMD is a number between 0 and 15 represented as a binary code
 t_{STEP} is the adjustment increment (see section 4.5)

3.3.4 Input Data Formats

The CH7015 supports 5 different multiplexed data formats, each of which can be used with a 1X clock latching data on both clock edges, or a 2X clock latching data with a single edge (rising or falling depending on the value of the MCP bit – rising refers to a rising edge on the XCLK signal, a falling edge on the XCLK* signal). The input data formats are (IDF[2:0]):

IDF	Description
0	12-bit multiplexed RGB input (24-bit color), (multiplex scheme 1)
1	12-bit multiplexed RGB input (24-bit color), (multiplex scheme 2)
2	8-bit multiplexed RGB input (16-bit color, 565)
3	8-bit multiplexed RGB input (15-bit color, 555)
4	8-bit multiplexed YCrCb input (24-bit color), (Y, Cr and Cb are multiplexed)

The input data format is shown in Figure 5 below. The Pixel Data bus represents a 12-bit or 8-bit multiplexed data stream, which contains either RGB or YCrCb formatted data. The input data rate is 2X the pixel rate, and each pair of Pn values (e.g., P0a and P0b) will contain a complete pixel encoded as shown in Table 10 through Table 13.

It is assumed that the first clock cycle following the leading edge of the incoming horizontal sync signal contains the first word (Pxa) of a pixel, if an active pixel was present immediately following the horizontal sync. This does not mean that active data should immediately follow the horizontal sync, however. When the input is a YCrCb data stream the color-difference data will be transmitted at half the data rate of the luminance data, with the sequence being set as Cb, Y, Cr, Y, where Cb0,Y0,Cr0 refers to co-sited luminance and color-difference samples and the following Y1 byte refers to the next luminance sample, per ITU-R BT.656 standards (the clock frequency is dependent upon the current mode, and is not 27MHz as specified in ITU-R BT.656). All non-active pixels should be 0 in RGB formats, and 16 for Y, 128 for Cr and Cb in YCrCb formats.

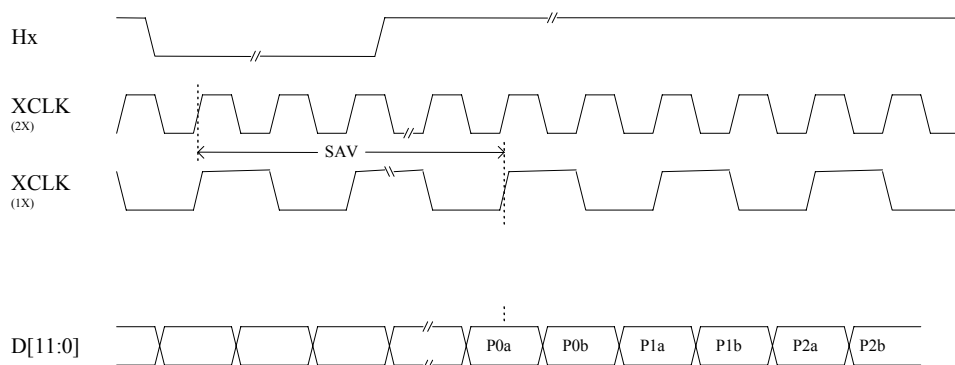


Figure 5: 12-bit Multiplexed Input Data Formats (IDFx = 0,1,2,3,4)

Table 10: Multiplexed Input Data Formats (IDF = 0, 1)

IDF = Format =		0 12-bit RGB				1 12-bit RGB			
Pixel #		P0a	P0b	P1a	P1b	P0a	P0b	P1a	P1b
Bus Data	D[11]	G0[3]	R0[7]	G1[3]	R1[7]	G0[4]	R0[7]	G1[4]	R1[7]
	D[10]	G0[2]	R0[6]	G1[2]	R1[6]	G0[3]	R0[6]	G1[3]	R1[6]
	D[9]	G0[1]	R0[5]	G1[1]	R1[5]	G0[2]	R0[5]	G1[2]	R1[5]
	D[8]	G0[0]	R0[4]	G1[0]	R1[4]	B0[7]	R0[4]	B1[7]	R1[4]
	D[7]	B0[7]	R0[3]	B1[7]	R1[3]	B0[6]	R0[3]	B1[6]	R1[3]
	D[6]	B0[6]	R0[2]	B1[6]	R1[2]	B0[5]	G0[7]	B1[5]	G1[7]
	D[5]	B0[5]	R0[1]	B1[5]	R1[1]	B0[4]	G0[6]	B1[4]	G1[6]
	D[4]	B0[4]	R0[0]	B1[4]	R1[0]	B0[3]	G0[5]	B1[3]	G1[5]
	D[3]	B0[3]	G0[7]	B1[3]	G1[7]	G0[0]	R0[2]	G1[0]	R1[2]
	D[2]	B0[2]	G0[6]	B1[2]	G1[6]	B0[2]	R0[1]	B1[2]	R1[1]
	D[1]	B0[1]	G0[5]	B1[1]	G1[5]	B0[1]	R0[0]	B1[1]	R1[0]
	D[0]	B0[0]	G0[4]	B1[0]	G1[4]	B0[0]	G0[1]	B1[0]	G1[1]

Table 11: Multiplexed Input Data Formats (IDF = 2, 3)

IDF = Format =		2 RGB 5-6-5				3 RGB 5-5-5			
Pixel #		P0a	P0b	P1a	P1b	P0a	P0b	P1a	P1b
Bus Data	D[11]	G0[4]	R0[7]	G1[4]	R1[7]	G0[5]	X	G1[5]	X
	D[10]	G0[3]	R0[6]	G1[3]	R1[6]	G0[4]	R0[7]	G1[4]	R1[7]
	D[9]	G0[2]	R0[5]	G1[2]	R1[5]	G0[3]	R0[6]	G1[3]	R1[6]
	D[8]	B0[7]	R0[4]	B1[7]	R1[4]	B0[7]	R0[5]	B1[7]	R1[5]
	D[7]	B0[6]	R0[3]	B1[6]	R1[3]	B0[6]	R0[4]	B1[6]	R1[4]
	D[6]	B0[5]	G0[7]	B1[5]	G1[7]	B0[5]	R0[3]	B1[5]	R1[3]
	D[5]	B0[4]	G0[6]	B1[4]	G1[6]	B0[4]	G0[7]	B1[4]	G1[7]
	D[4]	B0[3]	G0[5]	B1[3]	G1[5]	B0[3]	G0[6]	B1[3]	G1[6]

Table 12: Multiplexed Input Data Formats (IDF = 4)

IDF = Format =		4 YCrCb 8-bit							
Pixel #		P0a	P0b	P1a	P1b	P2a	P2b	P3a	P3b
Bus Data	D[7]	Cb0[7]	Y0[7]	Cr0[7]	Y1[7]	Cb2[7]	Y2[7]	Cr2[7]	Y3[7]
	D[6]	Cb0[6]	Y0[6]	Cr0[6]	Y1[6]	Cb2[6]	Y2[6]	Cr2[6]	Y3[6]
	D[5]	Cb0[5]	Y0[5]	Cr0[5]	Y1[5]	Cb2[5]	Y2[5]	Cr2[5]	Y3[5]
	D[4]	Cb0[4]	Y0[4]	Cr0[4]	Y1[4]	Cb2[4]	Y2[4]	Cr2[4]	Y3[4]
	D[3]	Cb0[3]	Y0[3]	Cr0[3]	Y1[3]	Cb2[3]	Y2[3]	Cr2[3]	Y3[3]
	D[2]	Cb0[2]	Y0[2]	Cr0[2]	Y1[2]	Cb2[2]	Y2[2]	Cr2[2]	Y3[2]
	D[1]	Cb0[1]	Y0[1]	Cr0[1]	Y1[1]	Cb2[1]	Y2[1]	Cr2[1]	Y3[1]
	D[0]	Cb0[0]	Y0[0]	Cr0[0]	Y1[0]	Cb2[0]	Y2[0]	Cr2[0]	Y3[0]

Table 13: Embedded Sync in Multiplexed Data Format (IDF=4)

IDF = Format =		4 YCrCb 8-bit							
Pixel #		P0a	P0b	P1a	P1b	P2a	P2b	P3a	P3b
Bus Data	D[7]	1	0	0	S[7]	Cb2[7]	Y2[7]	Cr2[7]	Y3[7]
	D[6]	1	0	0	S[6]	Cb2[6]	Y2[6]	Cr2[6]	Y3[6]
	D[5]	1	0	0	S[5]	Cb2[5]	Y2[5]	Cr2[5]	Y3[5]
	D[4]	1	0	0	S[4]	Cb2[4]	Y2[4]	Cr2[4]	Y3[4]
	D[3]	1	0	0	S[3]	Cb2[3]	Y2[3]	Cr2[3]	Y3[3]
	D[2]	1	0	0	S[2]	Cb2[2]	Y2[2]	Cr2[2]	Y3[2]
	D[1]	1	0	0	S[1]	Cb2[1]	Y2[1]	Cr2[1]	Y3[1]
	D[0]	1	0	0	S[0]	Cb2[0]	Y2[0]	Cr2[0]	Y3[0]

In this mode, the S[7:0] byte contains the following data:

S[6] = F = 1 during field 2, 0 during field 1

S[5] = V = 1 during field (frame) blank, 0 elsewhere

S[4] = H = 1 during EAV (synchronization reference at the end of active video)

0 during SAV (synchronization reference at the start of active video)

Bits S[7] and S[3:0] are ignored.

3.4 TV Output

3.4.1 Adaptive Flicker Filter

The CH7015 integrates an advanced 2-line, 3-line, 4-line, 5-line, 6-line and 7-line (depending on mode) vertical deflickering filter circuit to help eliminate the flicker associated with interlaced displays. This flicker circuit provides an adaptive filter algorithm for implementing flicker reduction with selections of high, medium or low flicker content for both luma and chroma channels (see register descriptions). In addition, a special text enhancement circuit incorporates additional filtering for enhancing the readability of text. These modes are fully programmable via serial port interface using the flicker filter register.

3.4.2 Color Burst Generation

The CH7015 allows the subcarrier frequency to be accurately generated from a 14.31818 MHz crystal oscillator, leaving the subcarrier frequency independent of the graphics pixel clock frequency. As a result, the CH7015 may be used with most VGA chips (with an appropriate digital interface) since the CH7015 subcarrier frequency can be generated without being dependent on the precise pixel rates of VGA controllers. This feature is important since even a $\pm 0.01\%$ subcarrier frequency variation is enough to cause some televisions to lose color lock.

In addition, the CH7015 has the capability to genlock the color burst signal to the VGA horizontal sync frequency, which enables a fully synchronous system between the graphics controller and the television. When genlocked, the CH7015 can stop “dot crawl” motion (for composite NTSC modes), thus eliminating the annoyance of moving borders. Both of these features are under programmable control through the register set.

3.4.3 NTSC and PAL Operation

Composite and S-Video outputs are supported in either NTSC or PAL format. The general parameters used to characterize these outputs are listed in Table 14 and shown in Figure 6 (see Figure 7 through Figure 14 for illustrations of composite and S-Video output waveforms).

ITU-R BT.470 Compliance

The CH7015 is predominantly compliant with the recommendations called out in ITU-R BT.470. The following are the only exceptions to this compliance:

- The frequencies of F_{sc} , F_h , and F_v can only be guaranteed in clock/sync master mode, not in clock/sync slave mode when the graphics device generates these frequencies.
- It is assumed that gamma correction, if required, is performed in the graphics device which establishes the color reference signals.
- All modes provide the exact number of lines called out for NTSC and PAL modes respectively, except mode 21, which outputs 800x600 resolution, scaled by 3:4, to PAL format with a total of 627 lines (vs. 625).
- Chroma signal frequency response will fall within 10% of the exact recommended value.
- Pulse widths and rise/fall times for sync pulses, front/back porches, and equalizing pulses are designed to approximate ITU-R BT.470 requirements, but will fall into a range of values due to the variety of clock frequencies used to support multiple operating modes.

Table 14: NTSC/PAL Composite Output Parameters

Symbol	Description	Level (mV)		Duration (uS)	
		NTSC	PAL	NTSC	PAL
A	Front Porch	287	300	1.49 - 1.51	1.48 - 1.51
B	Horizontal Sync	0	0	4.69 - 4.72	4.69 - 4.71
C	Breezeway	287	300	0.59 - 0.61	0.88 - 0.92
D	Color Burst	287	300	2.50 - 2.53	2.24 - 2.26
E	Back Porch	287	300	1.55 - 1.61	2.62 - 2.71
F	Black	340	300	0.00 - 7.50	0.00 - 8.67
G	Active Video	340	300	37.66 - 52.67	34.68 - 52.01
H	Black	340	300	0.00 - 7.50	0.00 - 8.67

For this table and all subsequent figures, key values are:

Note:

1. $R_{SET} = 140$ ohms; $V(ISET) = 1.235V$; 75 ohms doubly terminated load. R_{SET} is the resistor connected to the pin ISET.
2. Durations vary slightly in different modes due to the different clock frequencies used.
3. Active video and black (F, G, H) times vary greatly due to different scaling ratios used in different modes.
4. Black times (F and H) vary with position controls.

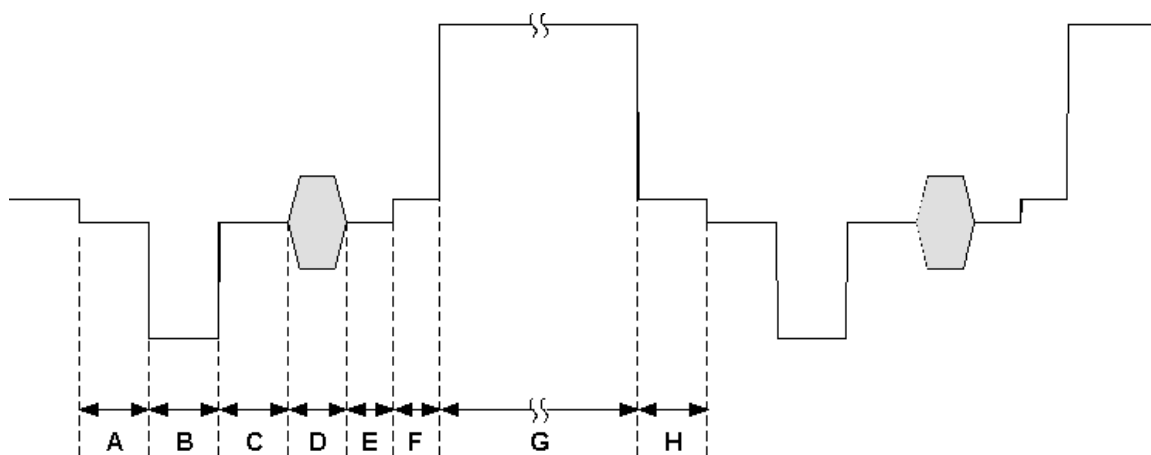


Figure 6: NTSC / PAL Composite Output

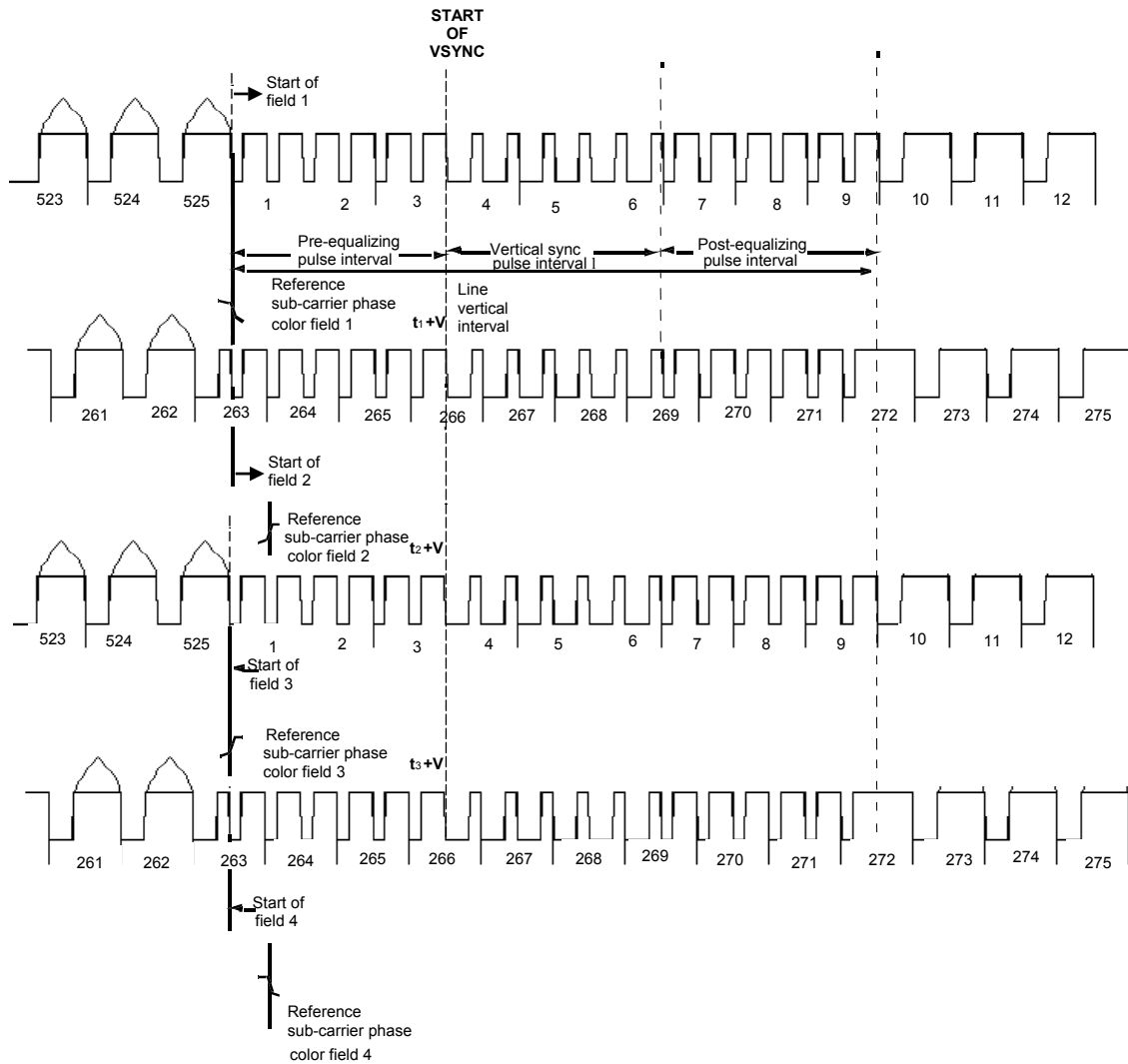


Figure 7: Interlaced NTSC Video Timing

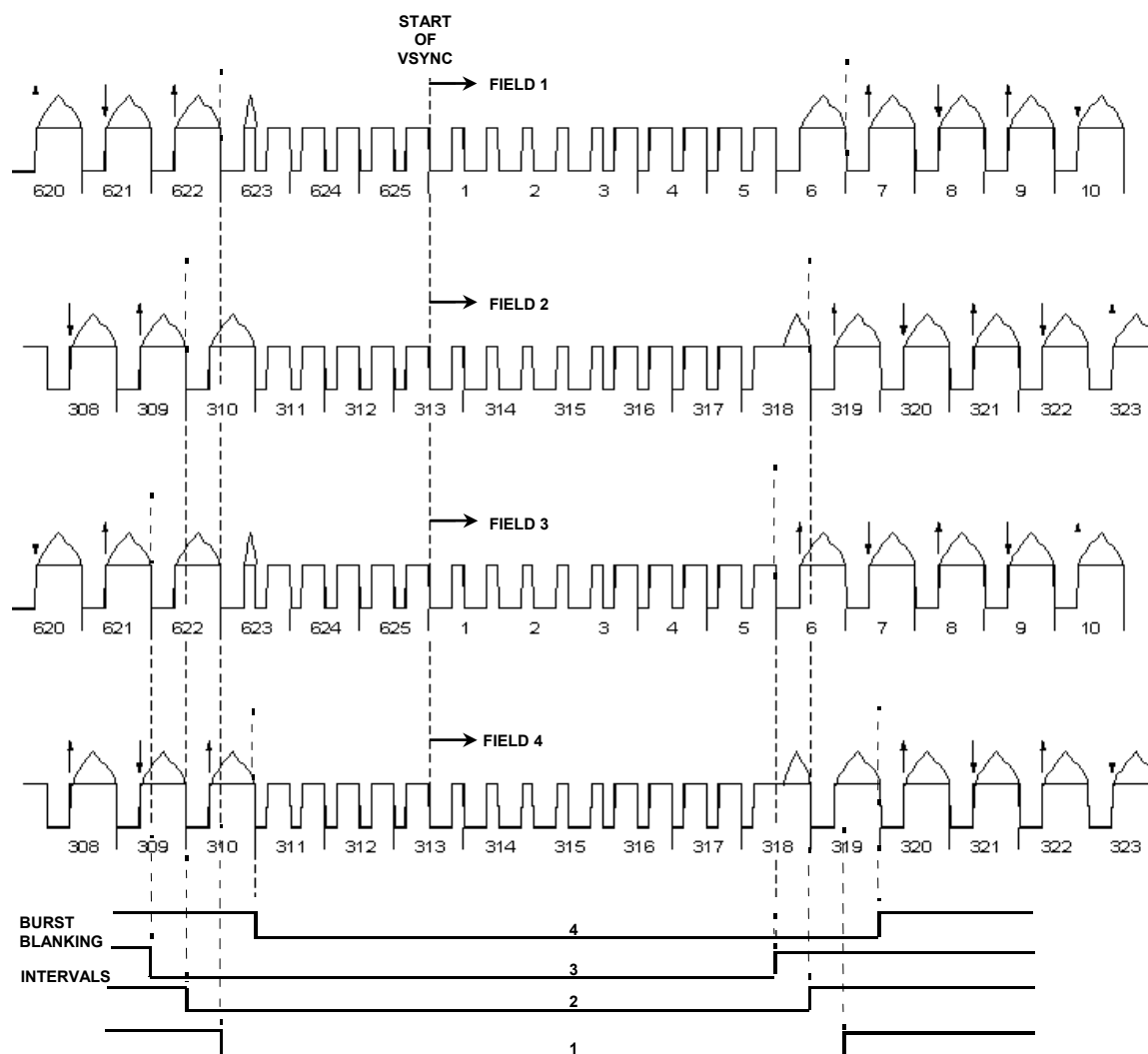


Figure 8: Interlaced PAL Video Timing

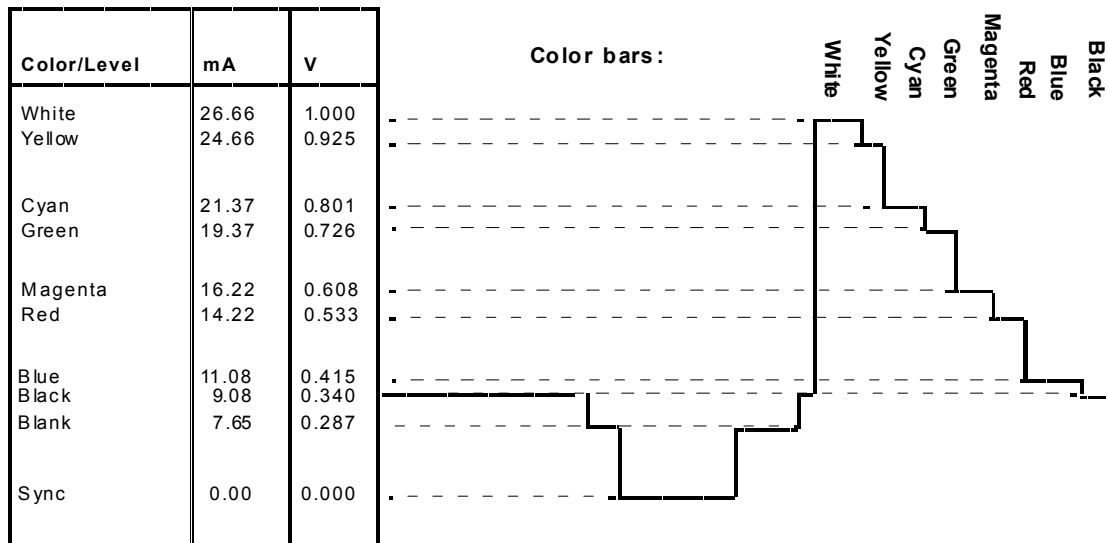


Figure 9: NTSC Y (Luminance) Output Waveform (DACG = 0)

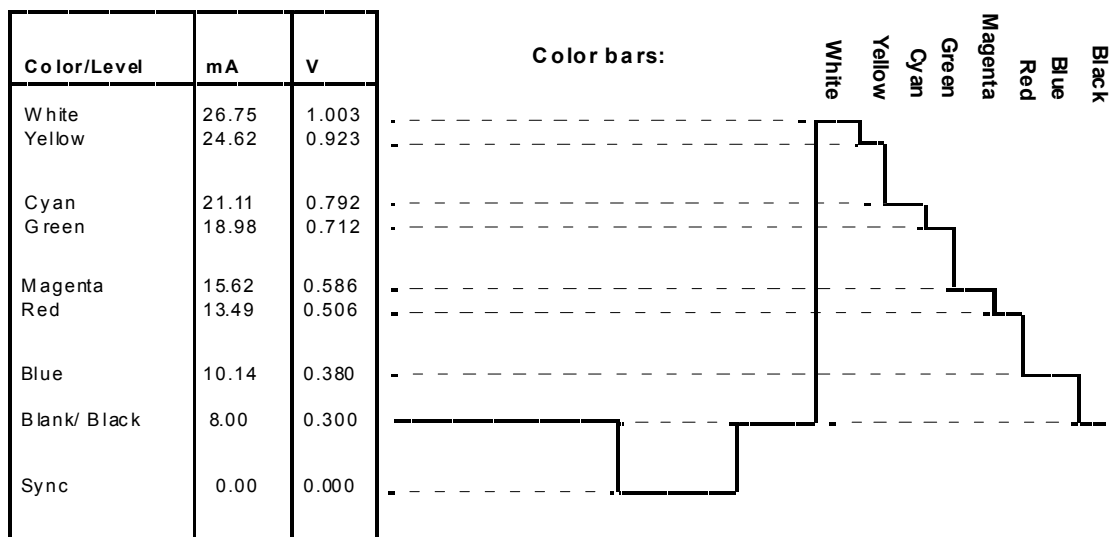


Figure 10: PAL Y (Luminance) Video Output Waveform (DACG = 1)

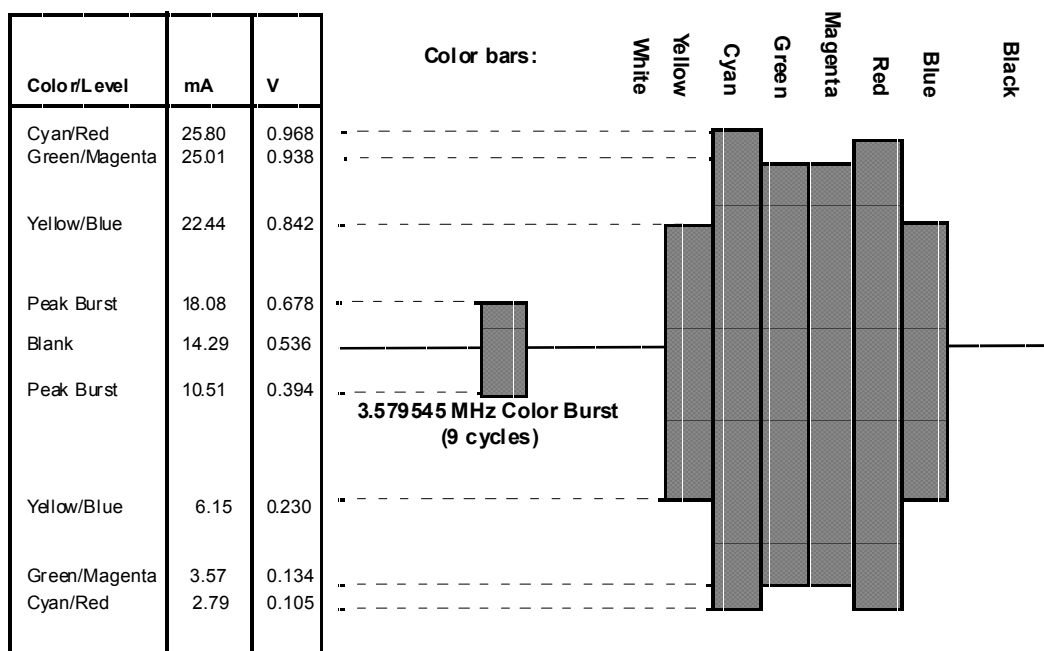


Figure 11: NTSC C (Chrominance) Video Output Waveform (DACG = 0)

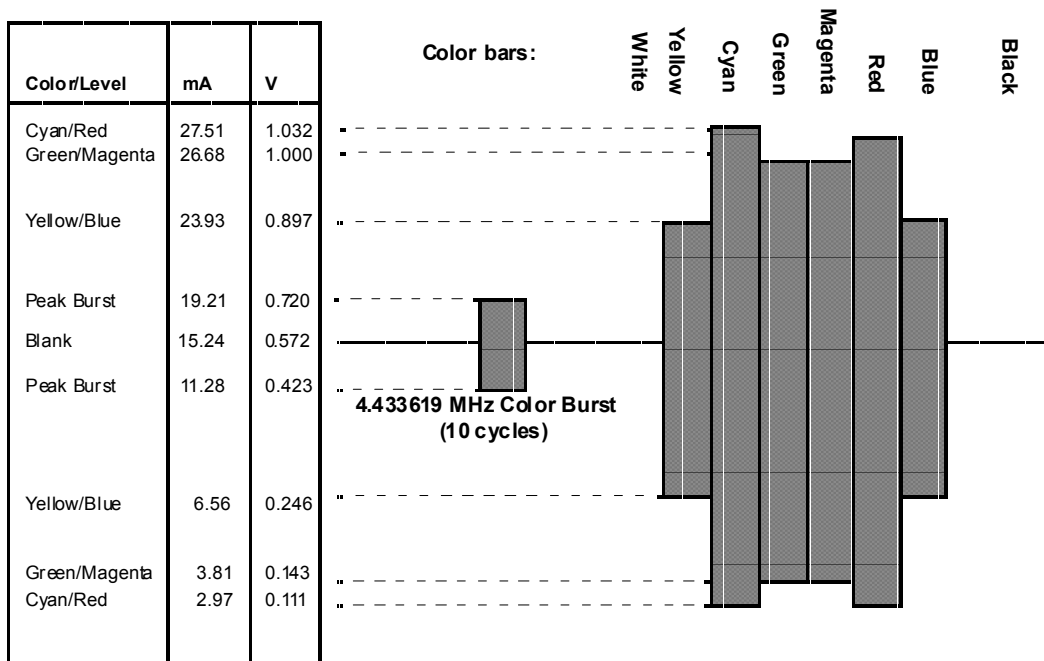


Figure 12: PAL C (Chrominance) Video Output Waveform (DACG = 1)

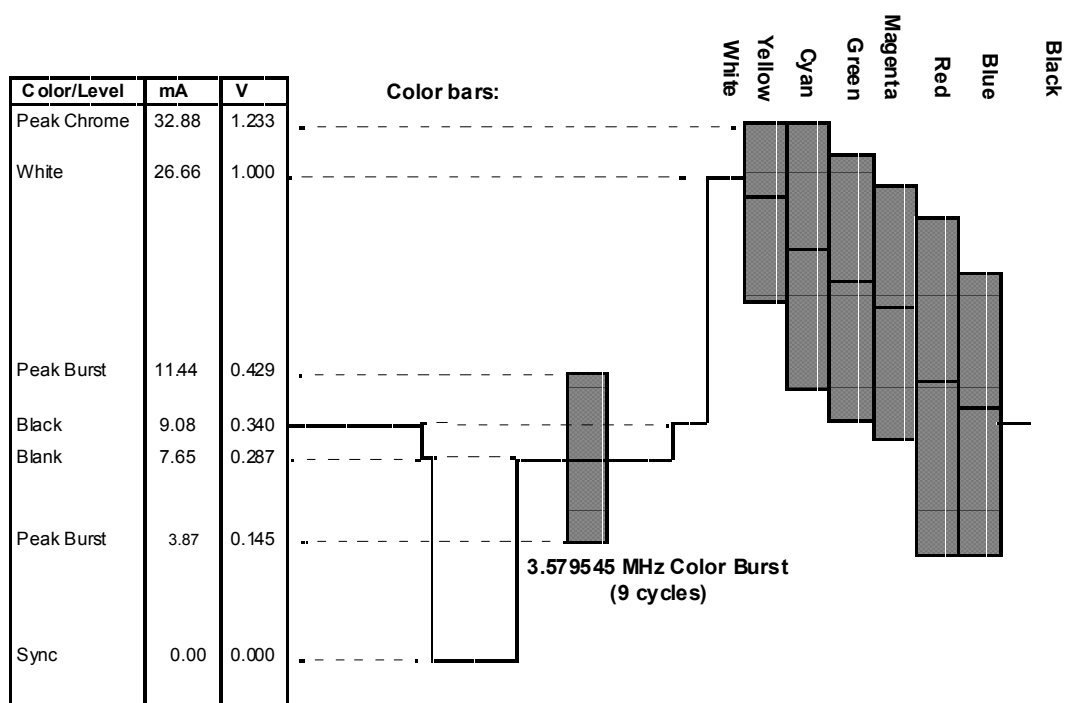


Figure 13: Composite NTSC Video Output Waveform (DACG = 0)

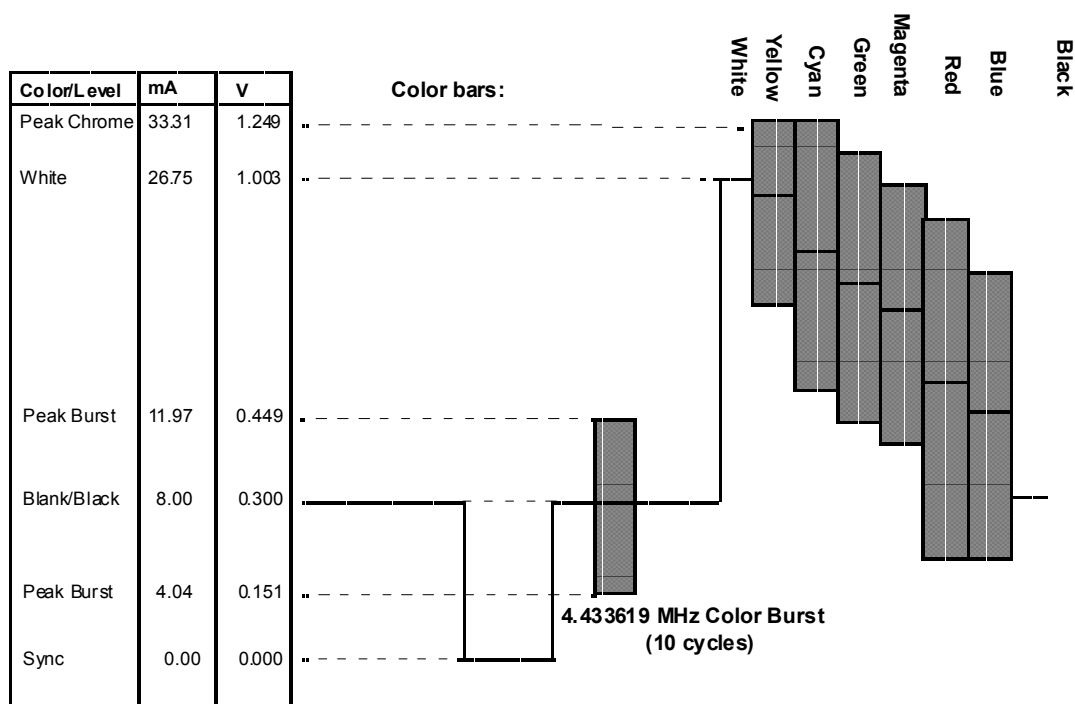


Figure 14: Composite PAL Video Output Waveform (DACG = 1)

3.5 Wide Screen Signaling

WSS algorithms, described by ITU-R.BT. 1119 and EIAJ CPR-1204, are used to instruct 16:9 TVs how to display 4:3 programs. Four WSS modes are supported by the CH7015. See the description for registers 31h – 33h for more details.

3.6 Median Filter

The CH7015 integrates a median filter in de-interlace mode. Instead of outputting an output line which is the simple average between two input lines the median filter analyses several pixels on adjacent lines and outputs the median of the pixel values. This results in reduced blurring in some situations.

The accuracy of the comparison of the pixel values can be controlled using the CMPLMT bit (bit 2 of register 35h). The median filter can be enabled using the MEDFEN bit (bit 0 of register 35h).

4.0 ELECTRICAL SPECIFICATIONS

4.1 Absolute Maximum Ratings

Symbol	Description	Min	Typ	Max	Units
	All power supplies relative to GND	-0.5		5.0	V
	Input voltage of all digital pins	GND – 0.5		VDD + 0.5	V
T _{SC}	Analog output short circuit duration		Indefinite		Sec
T _{AMB}	Ambient operating temperature	0		85	°C
T _{STOR}	Storage temperature	-65		150	°C
T _J	Junction temperature			150	°C
T _{VPS}	Vapor phase soldering (1 minute)			220	°C

Note:

- 1) Stresses greater than those listed under absolute maximum ratings may cause permanent damage to the device. These are stress ratings only. Functional operation of the device at these or any other conditions above those indicated under the normal operating condition of this specification is not recommended. Exposure to absolute maximum rating conditions for extended periods may affect reliability.
- 2) The device is fabricated using high-performance CMOS technology. It should be handled as an ESD sensitive device. Voltage on any signal pin that exceeds the power supply voltages by more than $\pm 0.5V$ can induce destructive latchup.

4.2 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Units
AVDD	PLL Power Supply Voltage	3.1	3.3	3.6	V
VDD	DAC Power Supply Voltage	3.1	3.3	3.6	V
DVDD	Digital Power Supply Voltage	3.1	3.3	3.6	V
VDDV	I/O Power Supply Voltage	1.1	1.8	3.6	V
R _L	Output load to DAC Outputs		37.5		Ω

4.3 Electrical Characteristics

(Operating Conditions: $T_A = 0^{\circ}\text{C} - 70^{\circ}\text{C}$, $V_{DD} = 3.3\text{V} \pm 5\%$)

Symbol	Description	Min	Typ	Max	Units
	Video D/A Resolution	10	10	10	bits
	Full scale output current		33.9		mA
	Video level error			10	%
I_{VDD}	Total supply current		200		mA
I_{VDDV}	VDDV (1.8V) current (15pF load on P-out)		4		mA
I_{PD}	Total Power Down Current		0.06		mA

4.4 Digital Inputs / Outputs

Symbol	Description	Test Condition	Min	Typ	Max	Unit
V_{SDOL}	SPD (serial port data) Output Low Voltage	$I_{OL} = 2.0 \text{ mA}$			0.4	V
V_{SPIH}	Serial Port (SPC, SPD) Input High Voltage		1.0		$V_{DD} + 0.5$	V
V_{SPIL}	Serial Port (SPC, SPD) Input Low Voltage		GND-0.5		0.4	V
V_{HYS}	Hysteresis of Inputs		0.25			V
V_{DATAIH}	D[0-11] Input High Voltage		$V_{ref} + 0.25$		$DV_{DD} + 0.5$	V
V_{DATAIL}	D[0-11] Input Low Voltage		GND-0.5		$V_{ref} - 0.25$	V
V_{MISCIH}	GPIOx, RESET* Input High Voltage	$DV_{DD} = 3.3\text{V}$	2.7		$V_{DD} + 0.5$	V
V_{MISCIL}	GPIOx, RESET* Input Low Voltage	$DV_{DD} = 3.3\text{V}$	GND-0.5		0.6	V
I_{MISCPU}	Pull Up Current (GPIO, RESET*)	$V_{IN} = 0\text{V}$	0.5		5.0	μA
V_{MISCOH}	GPIOx, BCO, H, V Output High Voltage	$I_{OH} = -0.4\text{mA}$	$DV_{DD} - 0.2$			V
V_{MISCOL}	GPIOx, BCO, H, V Output Low Voltage	$I_{OL} = 3.2\text{mA}$			0.2	V
$V_{P-OUTOH}$	P-OUT Output High Voltage	$I_{OH} = -0.4\text{mA}$	$V_{DDV} - 0.2$			V
$V_{P-OUTOL}$	P-OUT Output Low Voltage	$I_{OL} = 3.2 \text{ mA}$			0.2	V

Note :

V_{DATA} - refers to all digital data (D[11:0]), clock (XCLK, XCLK*) and sync (H, V) inputs. V_{MISC} - refers to GPIOx, and RESET* inputs and GPIOx, BCO/VSYN outputs and H, V when configured as outputs (SYO=1).

4.5 AC Specifications

Symbol	Description	Test Condition	Min	Typ	Max	Unit
f_{XCLK}	Input (XCLK) frequency		20		165	MHz
f_{XCLK}	Input (XCLK) frequency		20		27	MHz
DC_{XCLK}	Input (XCLK) Duty Cycle	$T_S + T_H < 1.2ns$	30		70	%
t_{XJIT}	XCLK clock jitter tolerance			2		ns
t_S	Setup Time: D[11:0], H, V and DE to XCLK, XCLK*	XCLK = XCLK* to D[11:0], H, V, DE = Vref	0.35			ns
t_H	Hold Time: D[11:0], H, V and DE to XCLK, XCLK*	D[11:0], H, V, DE = Vref to XCLK = XCLK*	0.5			ns
t_R	Pout, H and V (when configured as outputs) Output Rise Time (20% - 80%)	15pF load DVDD, VDDV = 3.3V			1.50	ns
t_F	Pout, H and V (when configured as outputs) Output Fall Time (20% - 80%)	15pF load DVDD, VDDV = 3.3V			1.50	ns
t_{STEP}	De-skew time increment		50		80	ps

4.6 Timing Information

4.6.1 Clock - Slave, Sync - Slave Mode

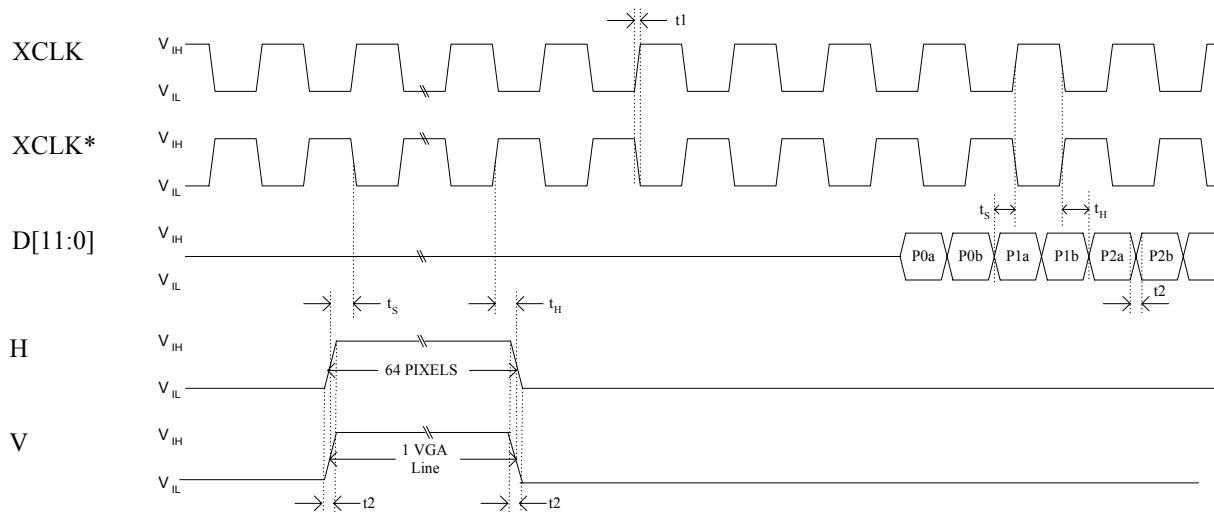


Figure 15: Timing for Clock - Slave, Sync - Slave Mode

Table 15: Timing for Clock - Slave, Sync - Slave Mode

Symbol	Parameter	Min	Typ	Max	Unit
t_s	Setup Time: D[11:0], H, V to XCLK, XCLK*	see section 4.5			
t_H	Hold Time: D[11:0], H, V to XCLK, XCLK*	see section 4.5			
t_1	XCLK & XCLK* rise/fall time w/15pF load		1		ns
t_2	D[11:0], H, V rise/fall time w/ 15pF load		1		ns

4.6.2 Clock - Master, Sync - Slave Mode

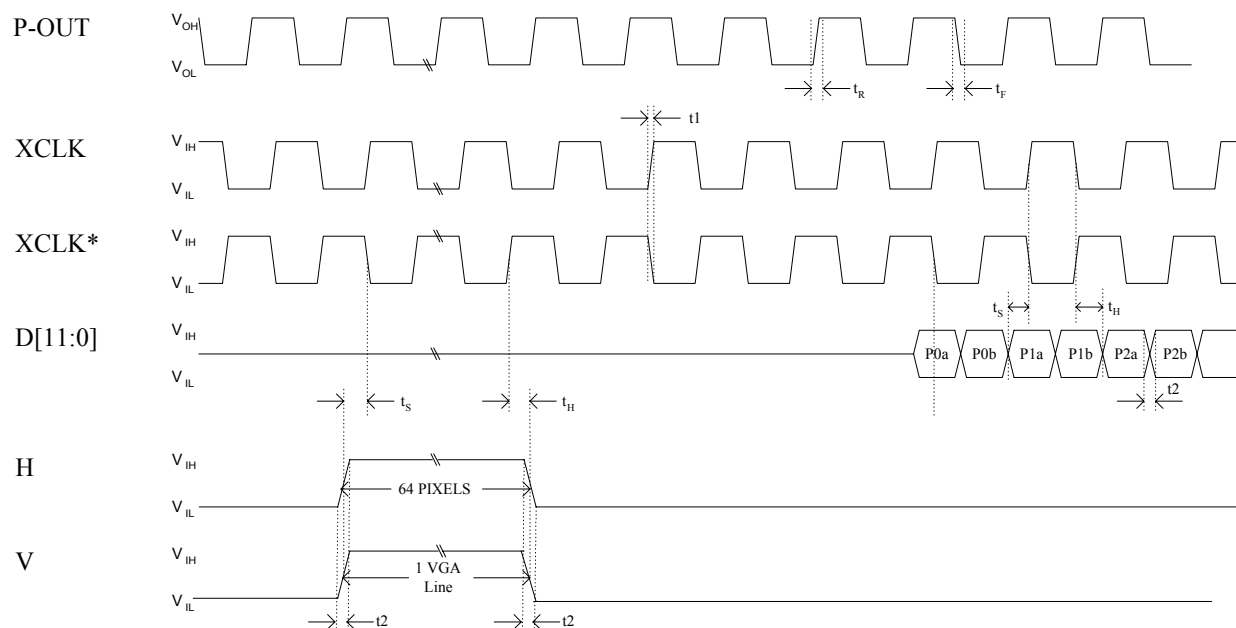


Figure 16: Timing for Clock - Master, Sync - Slave Mode

Table 16: Timing for Clock - Master, Sync - Slave Mode

Symbol	Parameter	Min	Typ	Max	Unit
t_s	Setup Time: D[11:0], H, V to XCLK, XCLK*	see section 4.5			
t_H	Hold Time: D[11:0], H, V to XCLK, XCLK*	see section 4.5			
t_R	Pout Output Rise Time	see section 4.5			
t_F	Pout Output Fall Time	see section 4.5			
t_1	XCLK & XCLK* rise/fall time w/15pF load		1		ns
t_2	D[11:0], H, V rise/fall time w/15pF load		1		ns

5.5.3 Clock - Master, Sync - Master Mode

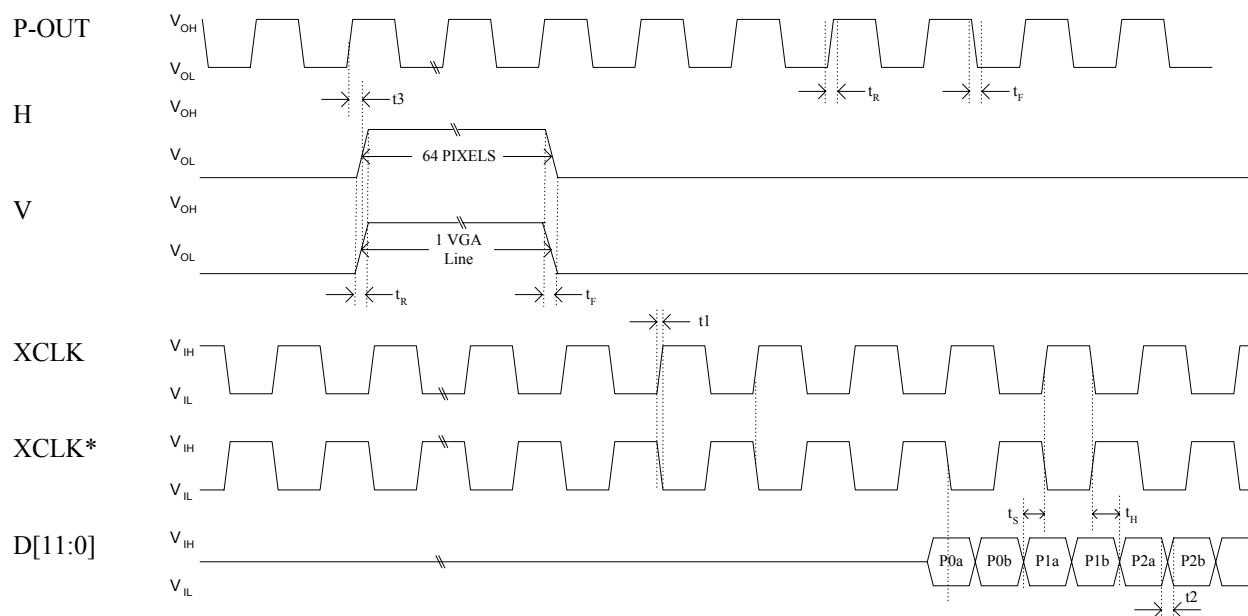


Figure 17: Clock - Master, Sync - Master Mode

Table 17: Timing for Clock - Master, Sync - Master Mode

Symbol	Parameter	Min	Typ	Max	Unit
t_S	Setup Time: D[11:0], H, V to XCLK, XCLK*	see section 4.5			
t_H	Hold Time: D[11:0], H, V to XCLK, XCLK*	see section 4.5			
t_R	Pout, H, V (when configured as outputs) Output Rise Time	see section 4.5			
t_F	Pout, H, V (when configured as outputs) Output Fall Time	see section 4.5			
t_1	XCLK & XCLK* rise/fall time w/15pF load		1		ns
t_2	D[11:0] rise/fall time w/15pF load		1		ns
t_3	Hold time: P-OUT to HSYNC, VSYNC delay		1.5		ns

5.0 PACKAGE DIMENSIONS

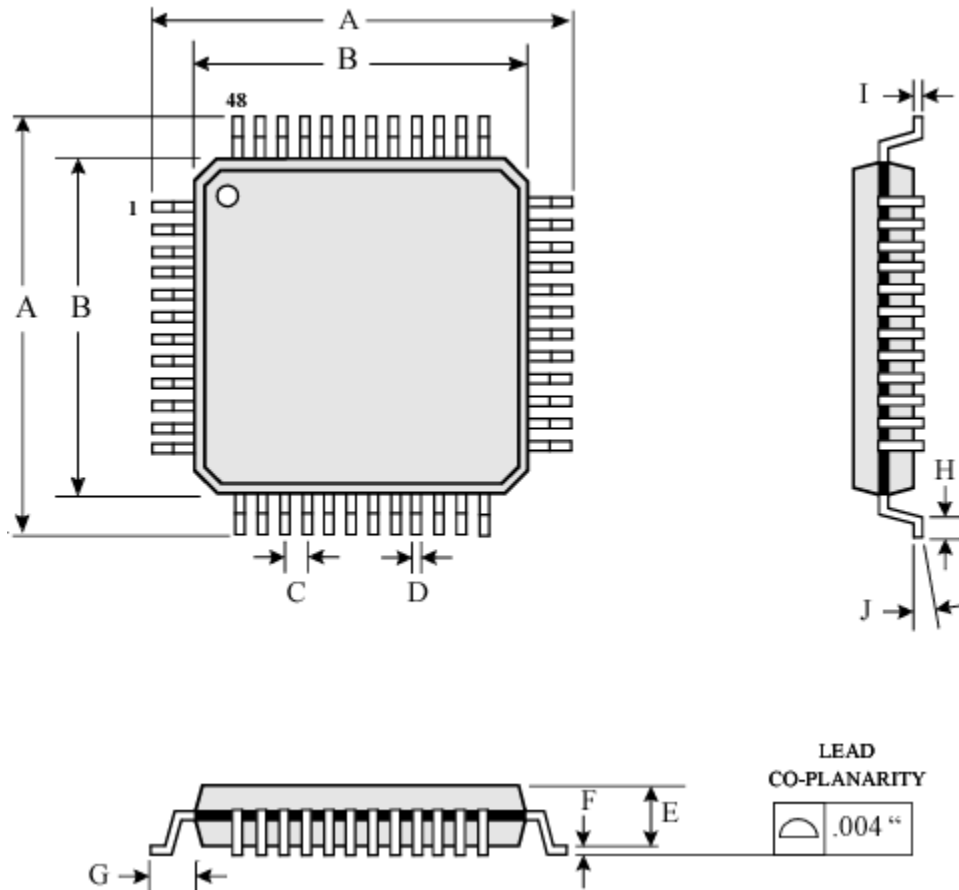


Table of Dimensions

No. of Leads		SYMBOL									
48 (7 X 7 mm)		A	B	C	D	E	F	G	H	I	J
Milli-meters	MIN	9	7	0.5	0.17	1.35	0.05	1.00	0.45	0.09	0°
	MAX				0.27	1.45	0.15		0.75	0.20	7°

Figure 18: 48 Pin LQFP Package

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ORDERING INFORMATION			
Part Number	Package Type	Number of Pins	Voltage Supply
CH7015A-T	LQFP	48	3.3V

Chrontel

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